Multi-stage events

1 $\frac{4}{9}$

2

a $\frac{1}{2}$ **b** $\frac{1}{2}$ **c** $\frac{219}{250}$ **d** $\frac{1}{10}$

e $\frac{321}{1000}$

3

a $\frac{1}{18}$ **b** $\frac{1}{36}$

4

a $\frac{1}{100}$ **b** $\frac{99}{100}$ **c** $\frac{1}{10\,000}$ **d** $\frac{1}{5000}$

e $\frac{1}{5000}$ **f** $\frac{4999}{5000}$

5

a $\frac{1}{25}$ **b** 1 **c** 0 **d** $\frac{6}{25}$

e $\frac{7}{25}$

6

a $\frac{1}{36}$ **b** $\frac{1}{36}$ **c** $\frac{1}{6}$ **d** $\frac{1}{36}$

7

a $\frac{5}{18}$ **b** $\frac{1}{6}$ **c** $\frac{7}{12}$

8

a $\frac{1}{36}$ **b** $\frac{1}{18}$ **c** $\frac{1}{12}$ **d** $\frac{1}{9}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Tables

9

a

	2	3	5	8
1	2	3	5	8
2	4	6	10	16
3	6	9	15	24
4	8	12	20	32
5	10	15	25	40
6	12	18	30	48

b 24

c

i $\frac{1}{4}$ ii $\frac{1}{12}$ iii $\frac{2}{3}$

10

a

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

b

i $\frac{1}{18}$ ii $\frac{1}{36}$ iii $\frac{1}{9}$ iv $\frac{1}{9}$

v $\frac{2}{3}$ vi $\frac{1}{6}$ vii $\frac{1}{18}$ viii $\frac{4}{9}$

c

i True ii False iii True iv False

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11

a

	7	9	12
2	9	10	11
3	11	12	13
4	14	15	16

b 9

c i $\frac{2}{9}$ ii $\frac{2}{9}$ iii $\frac{2}{9}$

12

a

	H	T
H	HH	HT
T	TH	TT

b $\frac{1}{4}$

c $\frac{3}{4}$

13

a

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

b $\frac{1}{4}$

c $\frac{3}{4}$

d $\frac{1}{3}$

14

a

Spinner	A	B	C
1	1,A	1,B	1,C
2	2,A	2,B	2,C
3	3,A	3,B	3,C

b 9

c $\frac{2}{9}$

d $\frac{7}{9}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

15

a

	W	X	Y	Z
1	1,W	1,X	1,Y	1,Z
2	2,W	2,X	2,Y	2,Z
3	3,W	3,X	3,Y	3,Z
4	4,W	4,X	4,Y	4,Z
5	5,W	5,X	5,Y	5,Z
6	6,W	6,X	6,Y	6,Z

b 24

c $\frac{1}{8}$

d $\frac{1}{6}$

e $\frac{11}{12}$

Lists

16

a $5283, 5238, 5823, 5832, 5328, 5382, 3285, 3258, 3825, 3852, 3528, 3582$

b 24

c $\frac{1}{4}$

d $\frac{1}{4}$

e $\frac{5}{12}$

17

a $689, 698, 869, 896, 968, 986$

b 6

c

i $\frac{1}{3}$ ii $\frac{1}{6}$ iii $\frac{1}{2}$

18

a $459, 495, 549, 594, 945, 954$

b

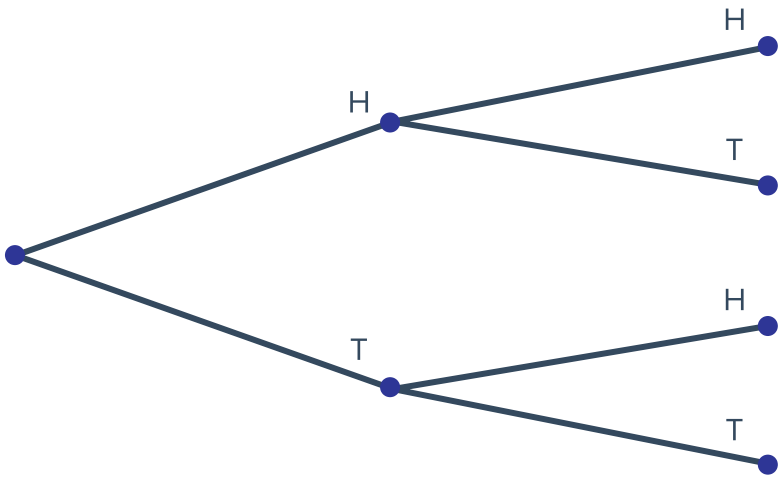
i $\frac{2}{3}$ ii $\frac{1}{3}$ iii $\frac{2}{3}$ iv $\frac{1}{3}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Tree diagrams

19

a

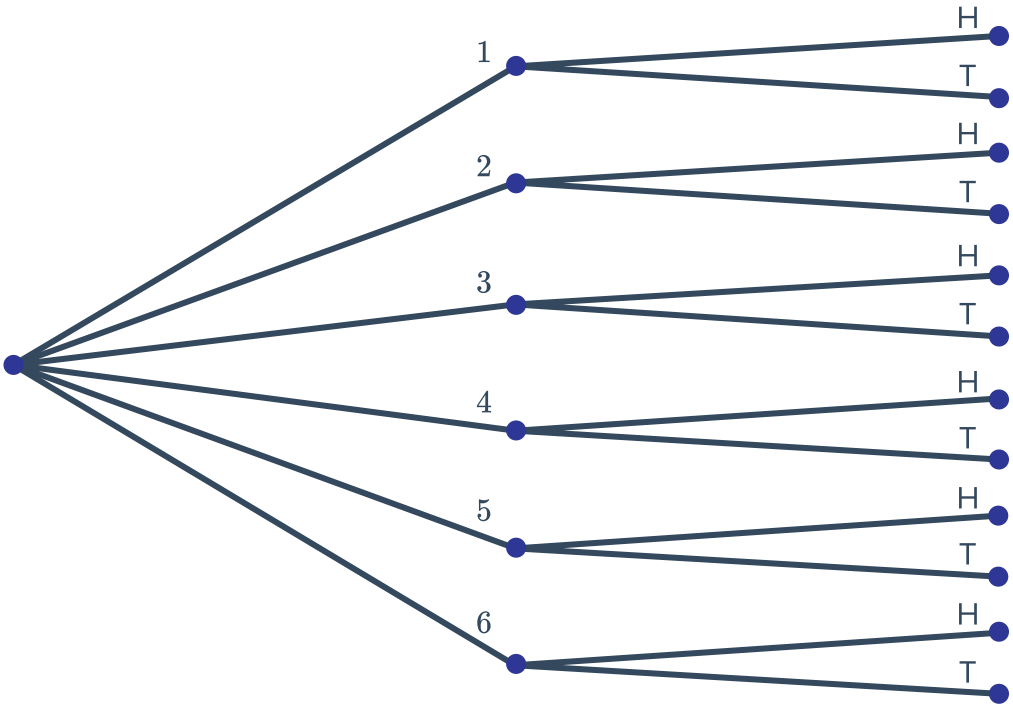


b $\frac{1}{4}$

c $\frac{3}{4}$

20

a



b $1\text{H}, 1\text{T}, 2\text{H}, 2\text{T}, 3\text{H}, 3\text{T}, 4\text{H}, 4\text{T}, 5\text{H}, 5\text{T}, 6\text{H}, 6\text{T}$

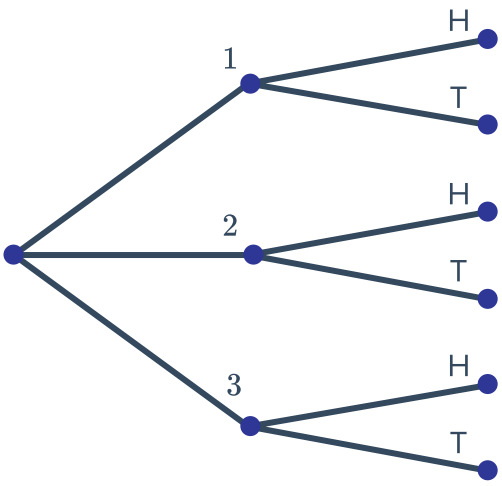
c $\frac{1}{2}$

d $\frac{5}{12}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21

a



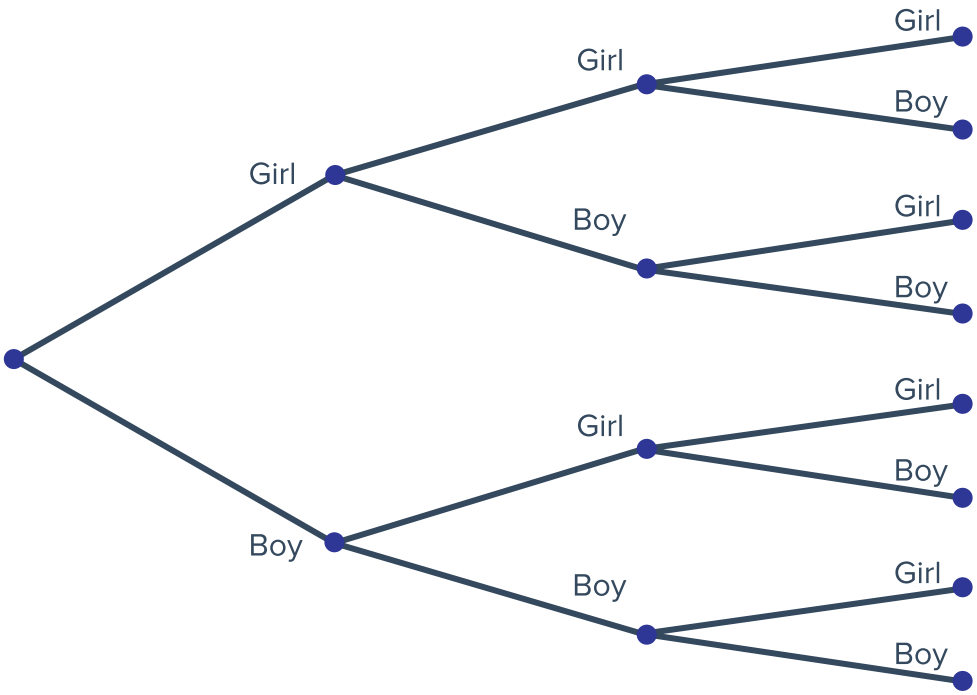
b $\frac{1}{6}$

c $\frac{1}{2}$

d $\frac{2}{3}$

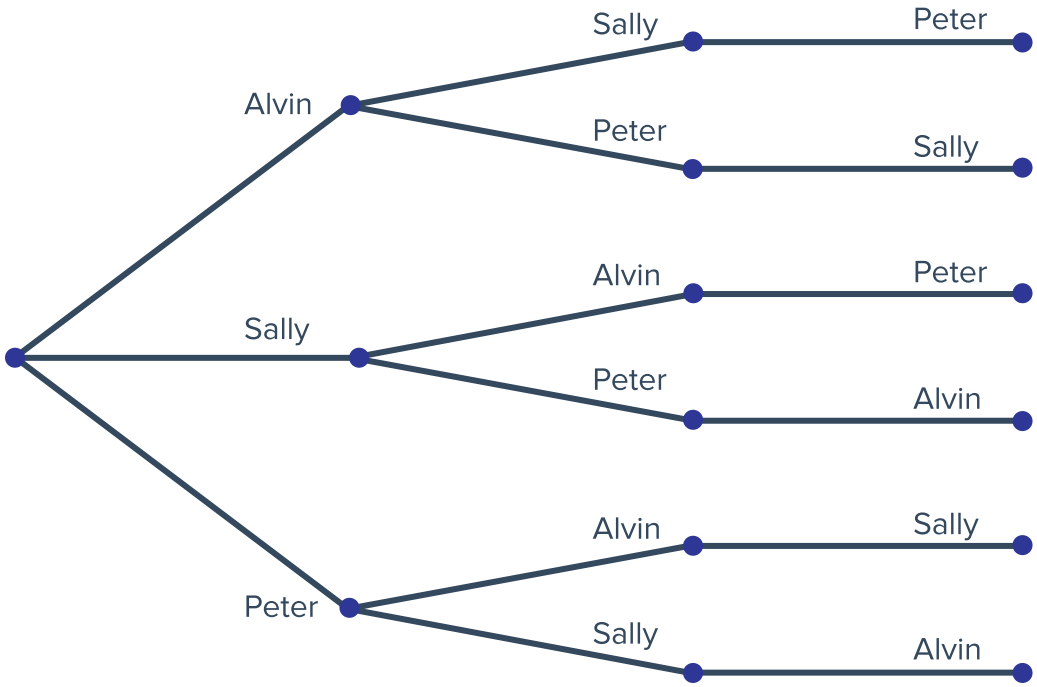
22

a



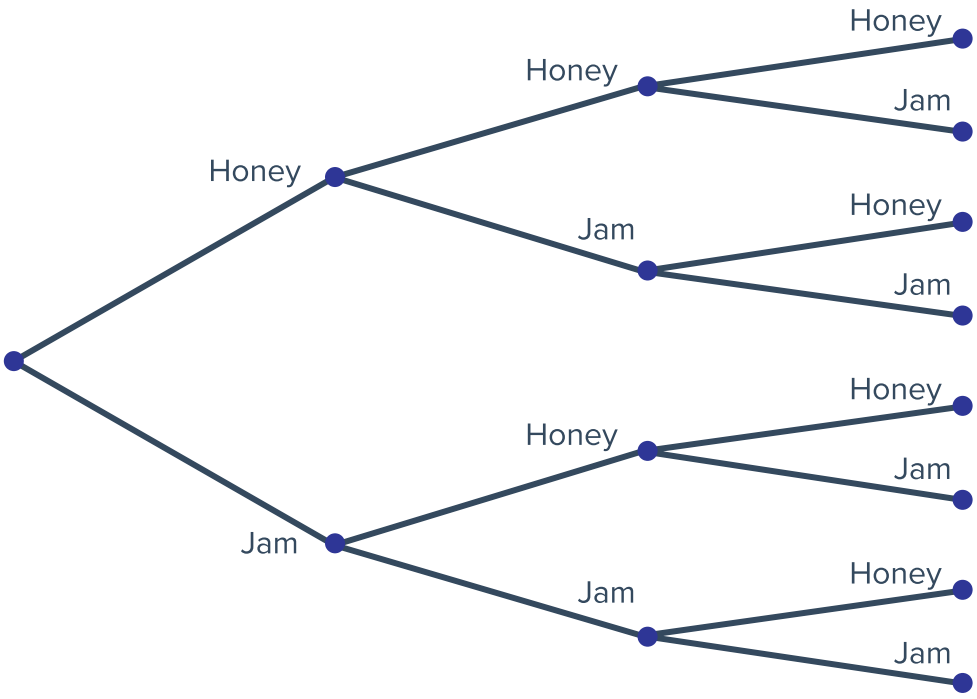
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

b



23

a



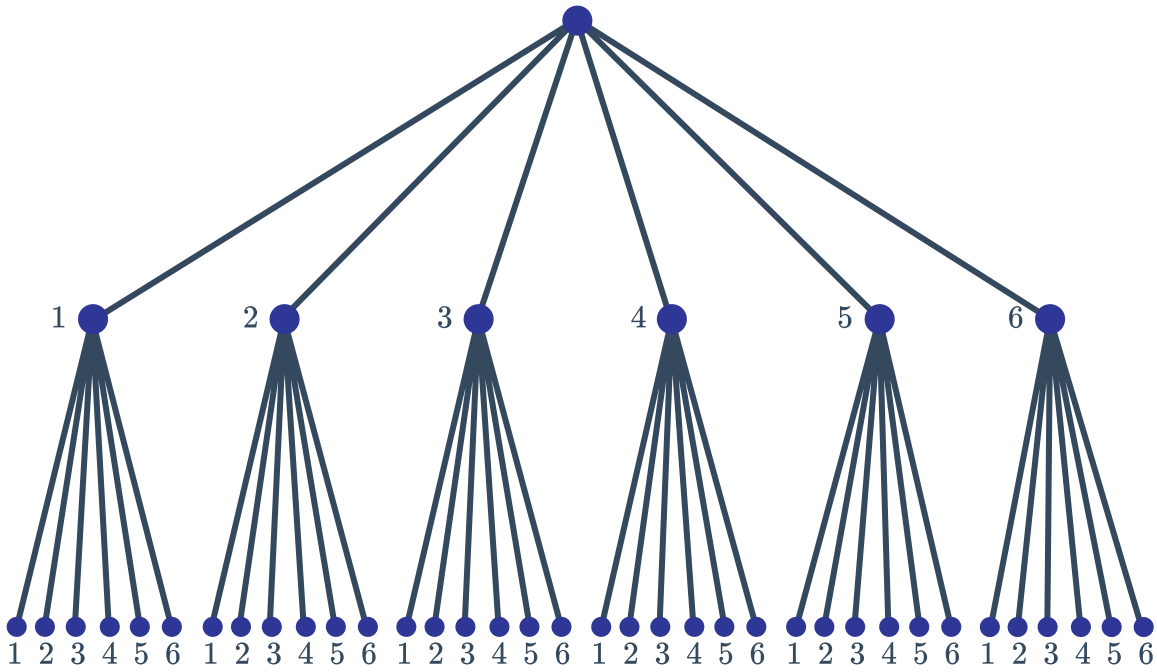
b $\frac{1}{2}$

c $\frac{1}{8}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

24

a

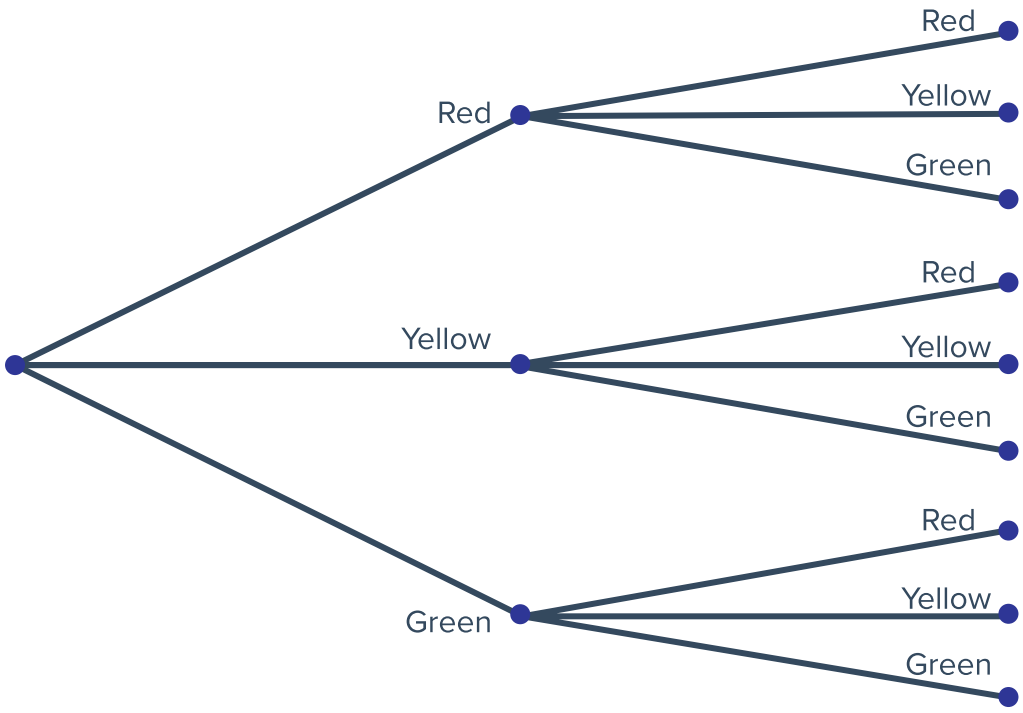


b

i $\frac{1}{36}$ ii $\frac{1}{6}$ iii $\frac{5}{6}$ iv $\frac{1}{4}$

25

a

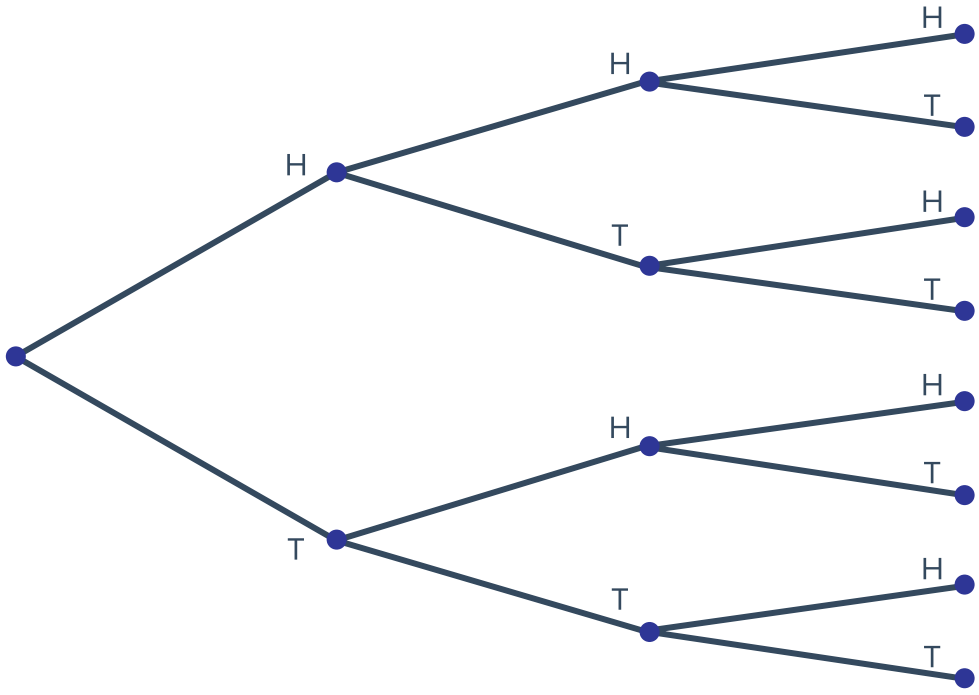


b $\frac{1}{9}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

26

a



b

i $\frac{7}{8}$

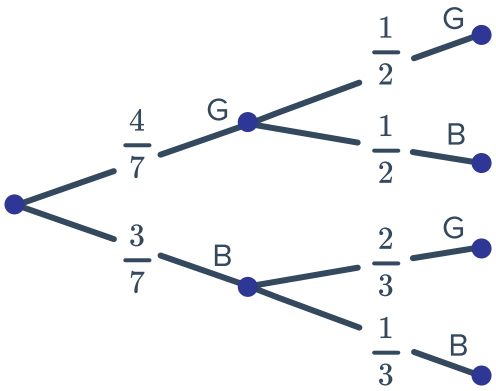
ii $\frac{1}{8}$

iii $\frac{1}{8}$

Probability trees

27

a



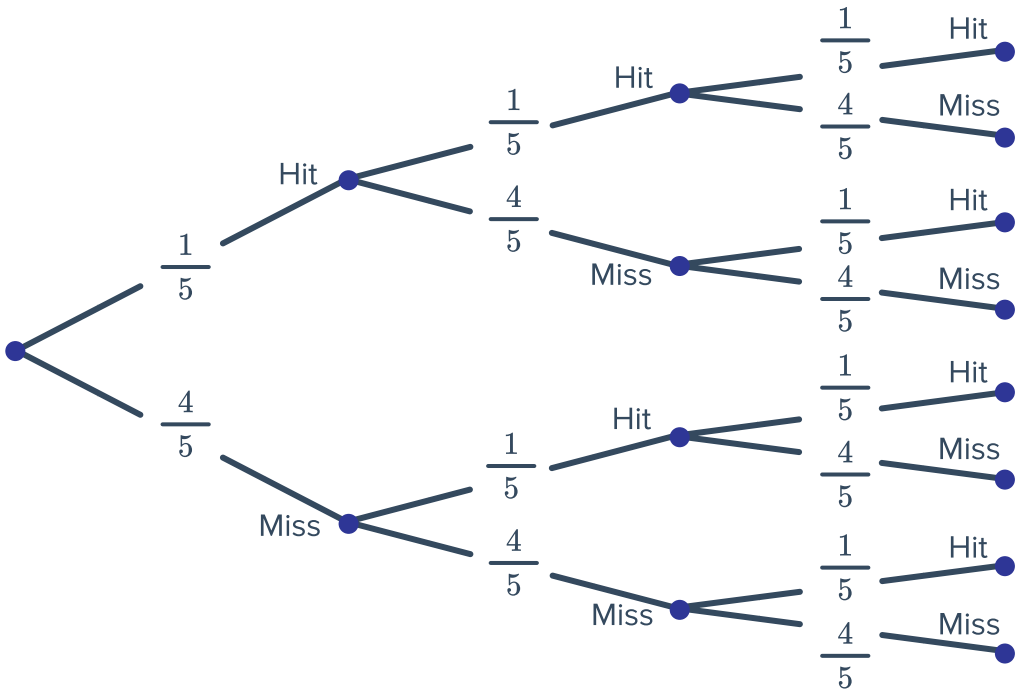
b $\frac{3}{7}$

c $\frac{2}{7}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

28

a



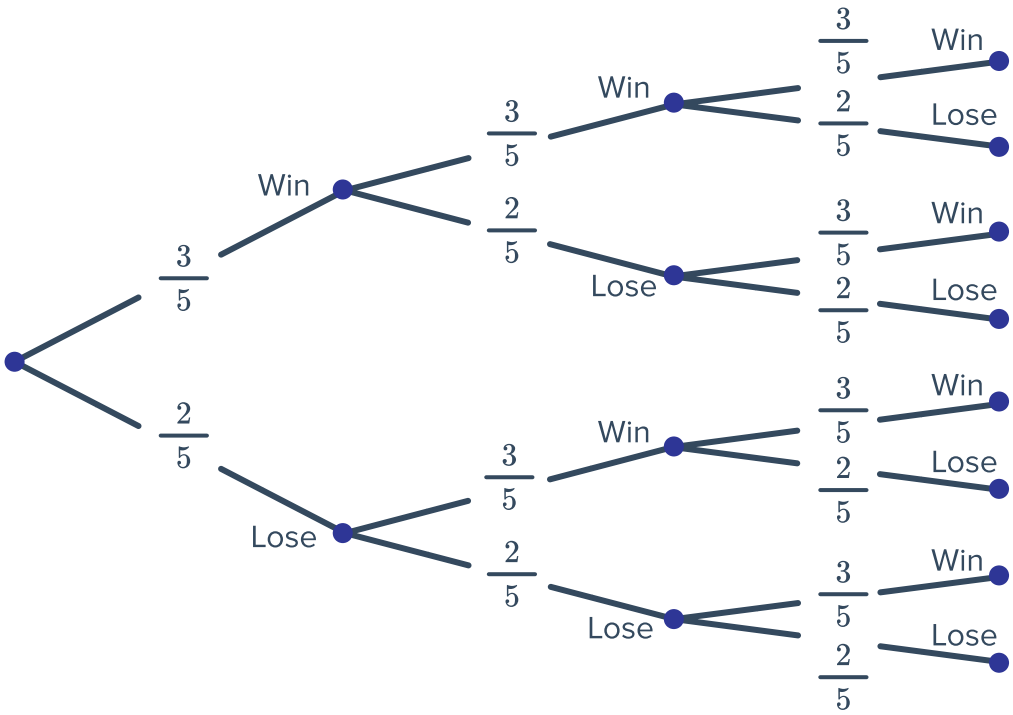
b $\frac{1}{125}$

c $\frac{124}{125}$

d $\frac{61}{125}$

29

a



b $\frac{27}{125}$

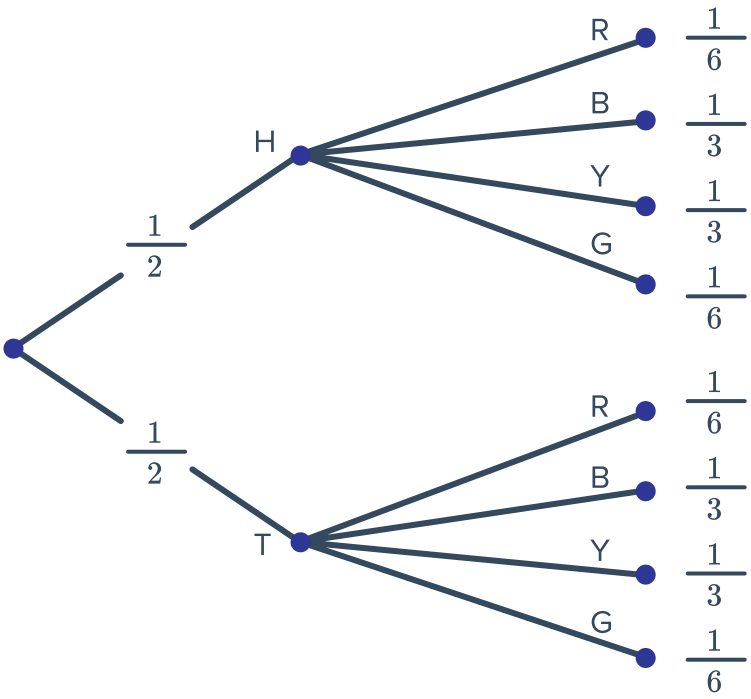
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c $\frac{8}{125}$

d $\frac{81}{125}$

30

a



b $\frac{1}{6}$

c $\frac{2}{3}$

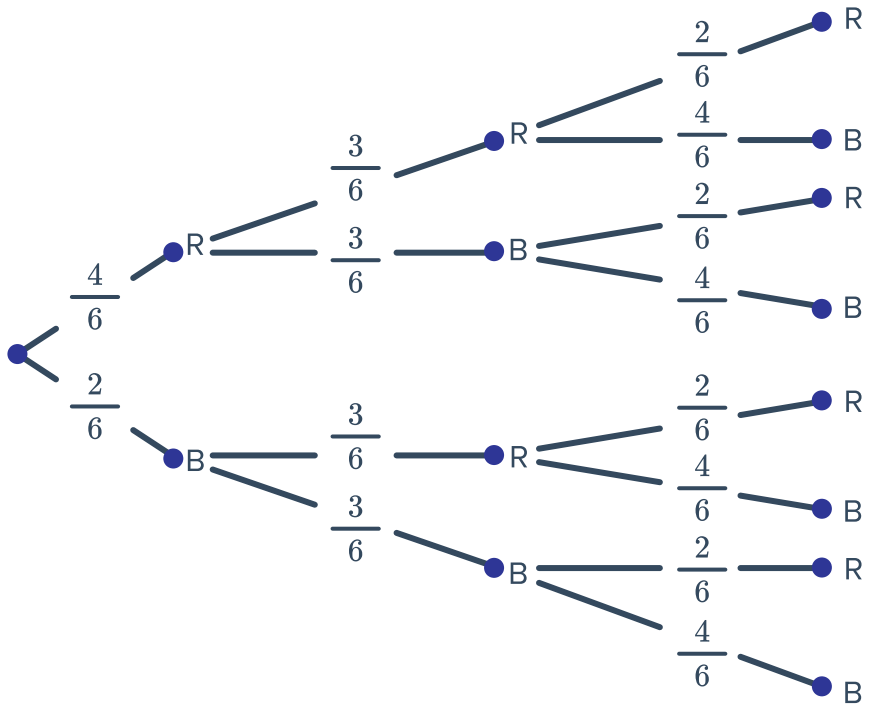
d $\frac{5}{12}$

e $\frac{5}{12}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31

a



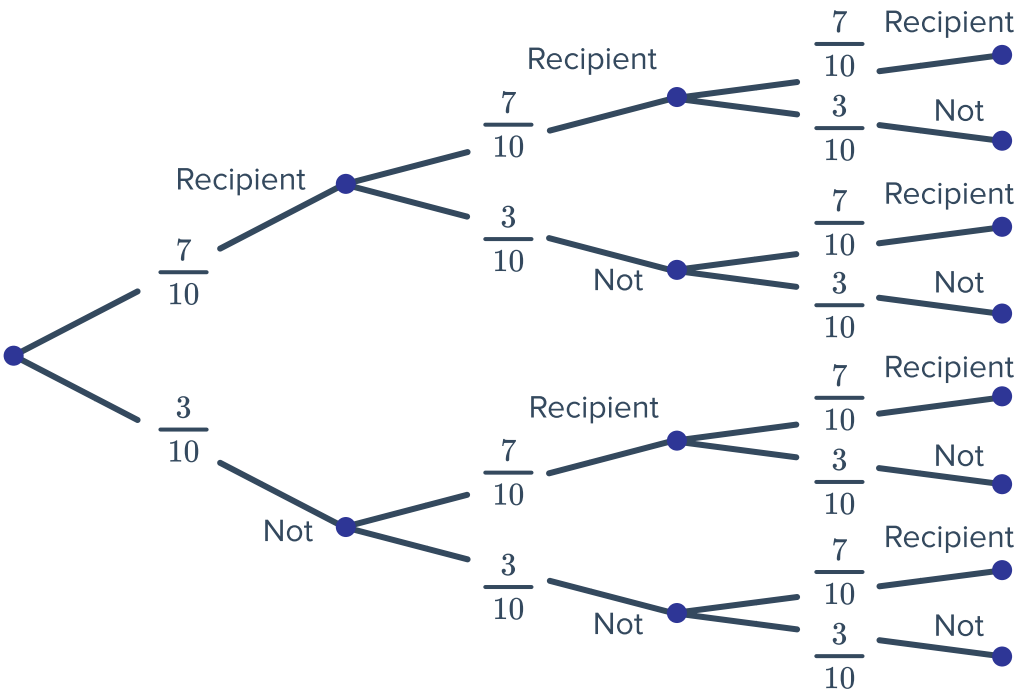
b $\frac{1}{9}$

c $\frac{1}{2}$

d $\frac{7}{18}$

32

a



b $\frac{973}{1000}$

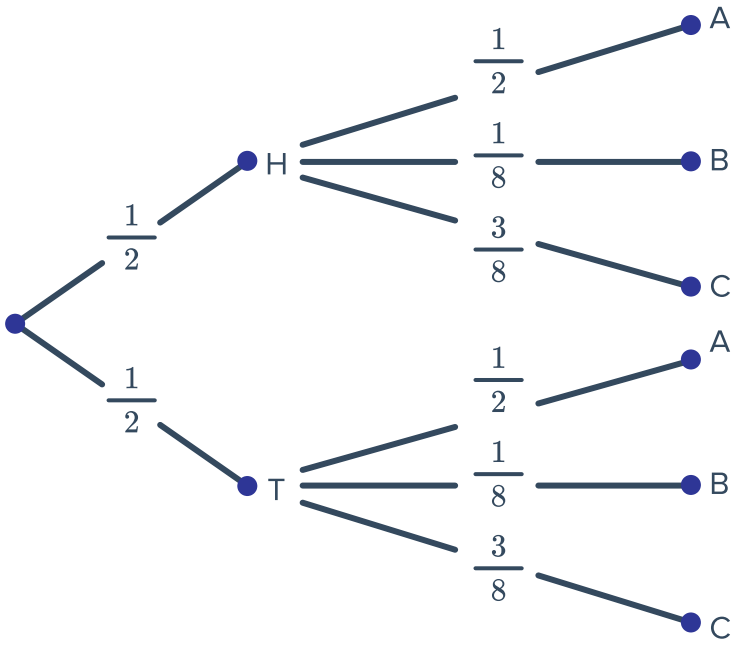
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c $\frac{657}{1000}$

d $\frac{63}{100}$

33

a



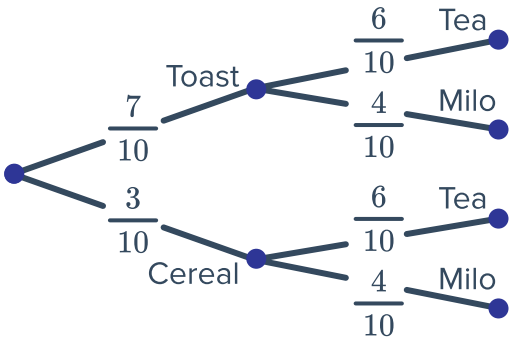
b $\frac{1}{4}$

c $\frac{3}{4}$

34

a

b 0.42



c 0.88

© ClipSAT: By Mr. Mohamed Abdallah +966597688647